

BARRANQUILLA, ColombiaWMO: **800280**Lat: **10.890N**Lon: **74.781W**Elev: **98**StdP: **14.64**Time Zone: **-5.00 (W05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.1	73.7	67.8	102.5	79.6	68.9	106.7	79.7	29.0	80.6	25.8	80.2	4.2	180	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.0	94.2	81.4	93.0	81.2	91.6	80.7	83.7	89.6	82.9	88.6	82.3	87.9	7.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.5	170.1	86.3	81.5	164.2	85.6	80.8	160.7	85.3	47.8	90.4	47.0	89.0	46.3	87.7	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.5	21.0	18.8	69.3	98.2	2.4	1.8	67.6	99.5	66.3	100.5	64.9	101.5	63.2	102.8	
(5)				67.2	85.4	2.3	1.4	65.5	86.4	64.2	87.2	62.9	88.0	61.2	89.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	81.9	82.0	82.9	84.1	84.3	84.1	83.9	84.1	83.7	82.8	82.7	82.8
	DBStd	2.15	1.50	1.73	1.84	1.82	2.12	1.99	2.01	2.15	2.42	2.30	2.08	1.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12146	988	897	1019	1024	1064	1022	1051	1056	1010	1016	981	1018
	CDD65	6671	523	477	554	574	599	572	586	591	560	551	531	553
	CDH74	69849	5016	4548	5477	6085	6573	6367	6485	6553	5994	5594	5436	5721
	CDH80	24663	1446	1354	1761	2171	2490	2441	2449	2564	2335	2010	1833	1809
Wind	WSAvg	8.4	11.4	12.3	12.3	10.5	7.1	6.5	7.4	6.9	5.8	5.4	6.2	9.2
Precipitation	PrecAvg	33.6	0.0	0.0	0.0	1.0	3.7	4.0	3.1	4.1	6.0	6.6	3.7	0.9
	PrecMax	63.7	0.3	0.3	0.5	5.6	11.3	14.1	14.1	15.8	11.8	14.3	15.2	6.6
	PrecMin	16.2	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.2	0.9	0.7	0.2	0.0
	PrecStd	11.4	0.0	0.1	0.1	1.3	2.6	2.9	3.0	2.9	2.7	3.2	3.0	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	91.8	93.4	94.9	95.0	94.7	94.6	95.1	95.3	93.6	92.5	91.8
		MCWB	78.6	78.6	80.0	81.0	81.5	81.5	80.9	81.3	82.3	82.7	81.3	80.4
	2%	DB	89.5	89.8	91.2	92.1	92.8	92.1	91.8	92.9	93.0	91.6	90.9	89.9
		MCWB	78.0	77.9	79.2	80.1	81.4	81.4	80.5	81.1	81.8	81.7	80.7	79.3
	5%	DB	87.9	88.1	89.4	91.0	91.2	91.1	91.0	91.2	91.1	89.8	89.4	88.5
		MCWB	77.7	77.5	78.6	79.6	80.9	81.1	80.3	80.8	81.3	81.0	80.3	79.0
	10%	DB	86.3	86.6	87.7	89.2	89.5	89.5	89.4	89.6	89.4	88.0	87.7	87.5
		MCWB	77.2	76.9	78.0	79.0	80.3	80.5	79.9	80.4	80.7	80.6	80.2	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.3	80.9	82.7	82.8	84.3	83.7	82.8	83.8	84.3	84.6	83.7	83.3
		MCDB	86.0	86.7	88.2	90.0	91.0	90.4	89.5	89.6	90.9	90.1	87.9	87.8
	2%	WB	80.0	79.7	80.9	81.5	82.7	82.6	81.8	82.6	83.1	83.2	82.3	81.6
		MCDB	85.4	85.9	87.0	88.2	88.9	89.0	88.5	88.7	89.0	88.3	87.1	86.4
	5%	WB	79.1	78.9	79.7	80.5	81.8	81.8	81.0	81.7	82.2	82.2	81.3	80.5
		MCDB	84.7	85.3	86.2	87.0	87.6	87.9	87.6	87.7	87.6	86.8	86.2	85.6
	10%	WB	78.2	78.0	78.8	79.7	80.9	81.0	80.3	80.9	81.4	81.3	80.5	79.6
		MCDB	83.9	84.4	85.4	86.0	86.5	86.9	86.4	86.7	86.5	86.0	85.4	84.5
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.0	12.3	13.0	13.0	13.0	13.0	13.7	14.1	13.3	12.4	11.6
		MCDBR	13.1	13.5	13.8	14.5	14.2	14.3	14.3	14.8	15.5	14.5	13.5	12.6
		MCWBR	5.4	5.4	5.2	5.2	5.6	5.9	5.5	5.9	6.6	6.6	5.9	5.5
	5% WB	MCDBR	12.1	12.5	13.3	13.6	13.5	13.5	13.4	13.9	14.6	13.8	12.7	12.0
MCWBR		5.8	5.9	5.7	5.4	5.9	6.2	5.8	6.2	7.0	7.2	6.5	6.0	
Clear-Sky Solar Irradiance	taub	0.398	0.431	0.466	0.511	0.473	0.461	0.454	0.452	0.449	0.413	0.401	0.401	
	taud	2.497	2.388	2.293	2.157	2.308	2.343	2.372	2.379	2.364	2.497	2.528	2.514	
	Ebn at Noon	288	282	272	258	263	263	266	270	273	284	285	285	
	Edh at Noon	34	39	44	50	42	40	39	39	40	35	33	33	
All-Sky Solar Radiation	RadAvg	1793	1920	1917	1855	1710	1772	1840	1826	1726	1558	1519	1634	
	RadStd	47	76	89	89	106	95	99	108	91	85	140	109	

Historical Trends

	DBAvg	Heating
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